

Arhaan Penwala

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EDUCATION

Vellore Institute of Technology

B.Tech Computer Science and Engineering (AI and Robotics), CGPA: 9.13

Chennai, India

2023 – 2027

EXPERIENCE

Software Development Engineer Intern

Dec 2025 – Feb 2026

BPO Integra India Private Limited

Remote

- Delivered full-stack product features into production across **2 sprint cycles**, building React.js UI components and Node.js/Express.js services on a MySQL-backed layer, validated through peer code review and unit testing
- Optimized query performance across **3+ RESTful API endpoints** while partnering with cross-functional teams in daily Agile standups to translate product requirements into production-ready components

AI Research Intern

May – June 2025

The Indian Hotels Company Limited (IHCL)

Mumbai, India

- Drove enterprise-wide selection of an AI chatbot partner for a Fortune 500 hospitality group, as measured by **35% identified cost savings** across 5+ vendors and 15+ use cases, by designing and building a structured cost-parametrization evaluation framework

PROJECTS

Cloud GCS | FastAPI, PostgreSQL/TimescaleDB, Redis, React/TypeScript, PySide6, Kotlin

Ongoing

- Architected a 4-component distributed ground control system (gateway, backend, frontend, SITL suite) for multi-drone fleet command
- Engineered horizontal scaling by deploying stateless FastAPI instances that share live and persistent state through Redis pub/sub and PostgreSQL/TimescaleDB
- Built a two-tier telemetry model with automatic bandwidth-aware rich/critical allocation per radio group, and shipped an Android gateway (Kotlin + Jetpack Compose wrapping a shared Python core via Chaquopy) fully interoperable with the desktop gateway

V Help | Flutter, Supabase (PostgreSQL), Firebase, Deno Edge Functions, Node.js

Ongoing

- Engineered a production campus super-app scaling to a **13,000+ student body**, adopted by **1,000+ users in the first hour**, unifying **16+ workflows** on a hybrid Supabase PostgreSQL (RLS) + Firebase Firestore backend across Android, iOS, and Web
- Deployed **33 serverless Deno Edge Functions** handling business logic, FCM push notifications, and cron jobs, reducing manual backend operations to zero

Junk-Wunk | 2nd Place, Hack-N-Droid | Flutter, AWS Cognito, S3, DynamoDB

2025

- Architected a role-based marketplace in Flutter with buyer and seller dashboards, replacing Firebase auth with AWS Cognito identity pools for secure, role-scoped session handling
- Implemented manual AWS Signature Version 4 (SigV4) signing in Dart for direct S3 media uploads, eliminating third-party SDK dependencies

Gait-Based Deepfake Detection | Siamese Verification, PyTorch, MediaPipe, GradCAM 2025 – 2026

- Built a deepfake detection pipeline extracting 33 MediaPipe skeletal landmarks into 78-dim gait signatures, powering a CNN + BiLSTM + Transformer hybrid with Siamese verification and GradCAM interpretability, achieving **94.95% AUC-ROC** via LOOCV
- Published the project dataset on IEEE DataPort (DOI: 10.21227/ngh5-b637); manuscript in preparation for peer-reviewed submission

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript/TypeScript, SQL, Kotlin

Frameworks & Libraries: React.js, Node.js, Express.js, FastAPI, Flutter, PyTorch, Scikit-learn, Pandas, XGBoost

Cloud & Databases: AWS (EC2, Cognito, S3), Firebase, Supabase, Docker, Git, PostgreSQL, MySQL, MongoDB, Redis

Core Concepts: Data Structures, Algorithms, Object-Oriented Programming, Software Engineering